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Cross section data sets for electron collisions with H_2, O_2, CO, CO_2, N_2O and H_2O

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Abstract. We review earlier cross section data sets for electron-collisions with H_2, O_2, CO, CO_2, H_2O and N_2O, updated here by experimental results for their electronic states. Based on our recent measurements of differential cross sections for the electronic states of those molecules, integral cross sections (ICSs) are derived by applying a generalized oscillator strength analysis and then assessed against theory (BE*f*-scaling [Y.-K. Kim, J. Chem. Phys. **126**, 064305 (2007)]). As they now represent benchmark electronic state cross sections, those ICSs for the above molecules are added into the original cross section sets taken from the data reviews for H_2, O_2, CO_2 and H_2O (the Itikawa group), and for CO and N_2O (the Zecca group).

1 Introduction

Quantification of the energy dependence of cross sections for fundamental electron excitation processes is essential for modelling and interpreting the behaviour of a large variety of naturally occurring and human-made phenomena. This includes space, our atmosphere and environment, chemical and electrical engineering, life and radiation sciences and fundamental areas of technology and energy. The demand for absolute, reliable, and comprehensive cross section sets for electron impact for various molecules is thus real and important. In general, one needs to consider a large variety of collision processes and one also needs to know the impact energy and angular dependence of the corresponding cross sections. In addition, benchmark data are required to guide development of theoretical models and computational schemes for calculating these cross sections. Electron-molecule collisions have been extensively studied for a long time, while cross sections have been reported in a wide variety of publications. The problem then is, firstly, to locate the numerical data and, secondly, to assess its accuracy. In these needs, however, it quickly becomes apparent that the available database on cross sections is still limited. This can be seen from the list of data compilations of electron-molecule collision cross sections as given by Itikawa [1], where the frequent review articles are cited chronologically for the database. Briefly, comprehensive compilations of cross section data for electron-molecule collisions have been published in the reviews from [2–5] and a volume of the Landolt-Börnstein series [6]. This includes the cross sections for ionization, electron-attachment, total scattering, elastic scattering, momentum transfer, and excitations of rotational, vibrational, and electronic states of more than seventy molecular species. For individual molecular species, like H_2 [7], O_2 [8], CO_2 [9] and H_2O [10], the Itikawa group published a series of data compilations in the *J. Phys. Chem. Ref. Data*, and for CO [3–5] and N_2O [3–5] the Zecca group reviewed a series of data compilations in *La Rivista del Nuovo Cimento*. Though, normally, the numerical data on differential cross sections (DCSs) are much too detailed to be summarized in a compact form, the Tanaka group

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has prepared three reports of numerical electron collision cross section databases on elastic DCS [11], vibrational excitation [12] and electronic excitations [13] for polyatomic molecules. Nonetheless, it is still usually claimed that the available electronic excitation cross section data remain unsatisfactory and are not as reliable compared with other total, elastic, vibrational and ionization cross sections.

Hence, in order to update a selection of the existing electron-molecule collision cross section data sets, we have reported measurements of differential cross sections for some of the electronic states in H_2 [14], O_2 [15], CO [16,17], CO_2 [18], H_2O [19], N_2O [20] and C_6H_6 [21]. Furthermore, a generalized oscillator strength analysis was employed in order to derive integral cross sections from the corresponding differential cross sections for those electronic excitation states. It was found that application of the energy and oscillator strength scaled Born cross section (BE*f*-scaling) approach, for calculating ICSs for dipole-allowed electronic-state transitions, can lead to a very accurate description for excitation of those states from threshold to a few keV.

In Section 2 of this report our measurement and analysis techniques from those recent studies [14–21] are described, along with a brief outline of the BE*f*-scaling theory and its procedures. In Section 3 our results and a discussion of these results are given and compared with the previous data in terms of the current database. Note we stress that this paper is not about assessing the quality of the available data bases, it simply adds new cross section information to those [3–5,7–10] data bases. Finally, the updated data sets for the six molecules of interest are summarized in 6 figures.

2 Experimental and analysis procedures

2.1 Experimental preliminaries

The apparatus employed in the present work has been described in detail elsewhere [22]. It consists of a crossed electron and molecular-beam configuration with a 180° monochromator, a single 180° energy analyzer and a computer-driven electron lens system. Overall the energy resolution was typically 40–60 meV (FWHM), as determined from the measured elastic peak in the energy loss spectrum. At the present resolution, the elastic and vibrational measurements include a small, unresolved rotational component. The experimental energy range of the recent studies was 15–200 eV, with the angular range of the DCS measurements being from −20° to +130°. The angular scale was calibrated to an accuracy of ±1.5°, as determined from the symmetry in the intensity profile of He 2^1P measured by changing the scattering angle of $\pm\theta$ against the nominal scattering angle at 0°. The energy scale was calibrated by reference to the resonance at 19.36 eV in He. For the incident energies of interest and over the energy loss range of interest, the transmission of the analyzer was corrected with our procedure for doing so being found in Kato et al. [14]. We employed the relative flow method [23] for both elimination of machine-dependent factors and normalization. The reference gas was He with the low-energy elastic cross section set compiled by Boesten and Tanaka [24] being used to set the absolute scale in each case. Experimental errors on the present inelastic DCSs are estimated to be about 18%−25%, including components due to the uncertainty in our analyzer transmission response, an uncertainty due to errors associated with the elastic normalization cross sections and uncertainties due to any fluctuations in the target density and/or the incident electron beam current during the measurements.

According to our program to provide the cross sections, we have studied the efficiency of the BE*f*-scaling approach in calculating reliable integral cross sections for electron impact excitation of dipole-allowed electronic states of many molecules. This work has been undertaken at all incident electron energies (E_0) above threshold to a few keV. The scaling, which was provided originally to estimate reliable total ionization cross sections of atoms and molecules by Kim and Rudd [25], has been extended to calculate the ICSs for dipole-allowed electronic-state transitions for atoms and molecules. To test the BE*f*-scaling approach we must therefore derive experimental ICSs from our measured DCSs. Our procedure for doing that is now briefly introduced as follows.

2.2 The generalized oscillator strength (GOS or G) fitting procedures

The values of [θ, DCS (θ)], for each electronic state, are transformed to (K^2, G_{expt}) using the standard formula [26]:

$$G_{expt}\left(K^2\right) = \frac{(E/R)k_i a_0}{4a_0^2 k_f a_0} K^2 DCS\left(E_0, \theta\right), \tag{1}$$

where k_i and k_f are the initial and final momenta of the incident electron, a_0 is the Bohr radius (0.529 Å), R is the Rydberg energy (13.6 eV), $G_{expt}\left(K^2\right)$ is the experimental generalized oscillator strength (GOS), and K^2 is the momentum transfer squared defined by:

$$K^2 = \left(k_i a_0\right)^2 + \left(k_f a_0\right)^2 - 2\left(k_i a_0\right)\left(k_f a_0\right)\cos\theta. \tag{2}$$

Vriens [27] proposed the following formula to represent the GOS for a dipole-allowed excitation:

$$G(x) = \frac{1}{(1+x)^6}\left[\sum_{m=0}^{\infty} \frac{f_m x^m}{(1+x)^m}\right], \tag{3}$$

where

$$x = \frac{K^2}{\alpha^2} \tag{4}$$

and

$$\alpha = \sqrt{\frac{B}{R}} + \sqrt{\frac{B-E}{R}}, \tag{5}$$

with B being the binding energy of the target electron being excited. In equation (3) the f_m are fitting constants

to be determined in a least-squares fit analysis of the experimental GOSs. The experimental optical oscillator strengths (OOSs) can now be extracted from the f_0 coefficient that was determined from the fit in each case. Finally, estimates of the ICS, at each energy, can be obtained from equations (6)–(8) using the standard formulae [28]:

$$ICS\left(E_0\right) = \frac{4\pi a_0^2}{E_0/R}\int_{K_{min}^2}^{K_{max}^2}\frac{G\left(K^2\right)}{E/R}d\ln\left(K^2\right), \qquad (6)$$

with

$$K_{min}^2 = 2\frac{E_0}{R}\left[1-\frac{E}{2E_0}-\sqrt{1-\frac{E}{E_0}}\right] \qquad (7)$$

and

$$K_{max}^2 = 2\frac{E_0}{R}\left[1-\frac{E}{2E_0}+\sqrt{1-\frac{E}{E_0}}\right]. \qquad (8)$$

Finally, having described our procedure for determining the experimental ICSs, we now briefly outline the method for calculating our theoretical ICSs.

2.3 Scaling methods

Scaled (plane wave) Born cross sections for dipole-allowed excitations – *f*-scaling, BE-scaling and BE*f*-scaling are described in detail by Kim [26]. Hence only a brief description is given here for the BE*f*-scaling approach. It is well known that the Born approximation gives fairly well the excitation cross sections for optically allowed states at high collision energies. Note that the Born cross sections are not only subject to approximations in the collision theory part, but also depend on the accuracy of the wave functions used for the initial and final states of the target molecules. Modifying the Born result, Kim originally developed a new, simple, method of calculation for the excitation cross section of atoms over the entire range of collision energy [29]. He then extended this to the case of a molecular target. It is in that form called the BE*f*-scaled Born method [26]. The BE*f*-scaled Born cross section σ_{BEf} is given by:

$$\sigma_{BEf}(E_0) = \frac{f_{accur}}{f_{Born}}\frac{E_0}{E_0+B+E}\sigma_{Born}(E_0), \qquad (9)$$

where E_0 is the energy of the incident electron, B is the binding energy of the electron being excited and E is the excitation energy. The optical oscillator strength is denoted by f. The quantity f_{accur} is an accurate value obtained from experiments, or calculations with accurate wave functions, and f_{Born} is the value from the calculation with the wave function employed in determining the unscaled Born cross section σ_{Born}. The *f*-scaling process has the effect of replacing the wave function used for σ_{Born} with an accurate one. Note that the BE-scaling corrects the deficiency of the Born approximation at low E_0, without losing its well-known validity at high E_0. In this report, it is found that the BE*f*-scaled Born method gives fairly good values for the cross section over the entire collision energy range investigated. In particular we specifically illustrate this for H_2 [14], O_2 [15] CO [16,17], CO_2 [18], H_2O [19], N_2O [20] and we also draw the attention of the reader to our work on C_6H_6 [21]. Thus, once the Born cross section is obtained theoretically or experimentally, a fairly reasonable cross section can be produced at any incident energy with the BE*f*-scaling approach. Note that exceptions to this general statement have only been found in those cases where resonance effects, due to the temporary capture of the incident electron by the target, and contamination from an accidentally degenerate or near-degenerate triplet state, have arisen.

3 Results and discussion

A typical energy loss spectrum for electron impact excitation of the relevant electronic states is shown for H_2 in Figure 1, O_2 in Figure 3, CO in Figure 5, CO_2 in Figure 7, N_2O in Figure 9, and H_2O in Figure 11. In parallel with each energy loss spectrum, a corresponding set of the updated cross sections is shown in Figures 2, 4, 6, 8, 10 and 12, respectively. Note that a full description of the new integral cross sections for each molecule can be found in the original papers [14–20]. Hence only a brief discussion of the more important details are given here in relation to updating the database, for each species (see below).

3.1 H_2

For H_2, electron energy loss (ΔE) spectroscopy (EELS) spectra were measured at various incident energies (E_0) in the range 40 to 200 eV, for the scattered electron angle (θ) range from 3.5° to 130°. In Figure 1 a typical EELS spectrum for H_2 is plotted for the $B^1\Sigma_u^+$ and $C^1\Pi_u$ excitation over the energy-loss range from 11.3 eV to 16.7 eV at $E_0 = 200$ eV and $\theta = 4.9°$. Here the complication in the data analysis is that there are many overlapping vibrational sublevels of the respective electronic states in the ∼11.3 to 16.7 eV energy-loss range. As a consequence, the various $B^1\Sigma_u^+$ and $C^1\Pi_u$ contributions to the energy-loss spectra had to be spectrally deconvolved. For this purpose we adapted the approach outlined in detail in Campbell et al. [30], which we modify here to incorporate the relevant energies and Franck-Condon factors from Wrkich et al. [31] and from the international atomic energy agency (IAEA) website[1].

The updated cross section data set for H_2 is shown in Figure 2 with our ICSs for the H_2 $^1\Sigma_u^+$ and $^1\Pi_u$ electronic excitations assessed against our BE*f*-scaling computation for the energy range from threshold to 1000 eV, as derived from the calculated result of Kim [26]. The ICSs are shown as the sum of the vibrational manifold for the sublevels included in each individual excitation for the $B^1\Sigma_u^+$ and $C^1\Pi_u$ states. Within the paradigm of the BE*f*-scaling

[1] http://www-amdis.iaea.org/data/INDC-457/

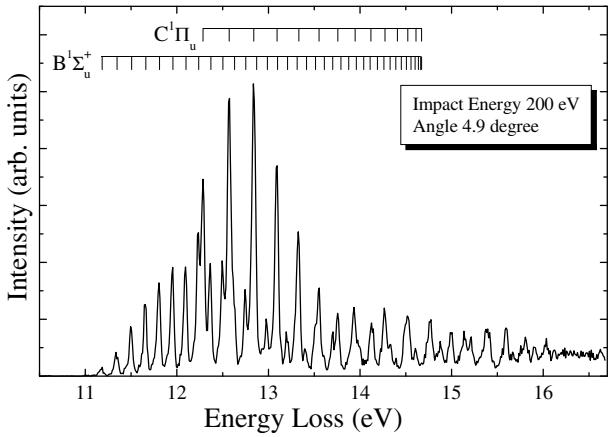


Fig. 1. Typical electron energy loss spectrum, $\Delta E \sim 11.3$–16.7 eV, for electronic state excitation in H_2.

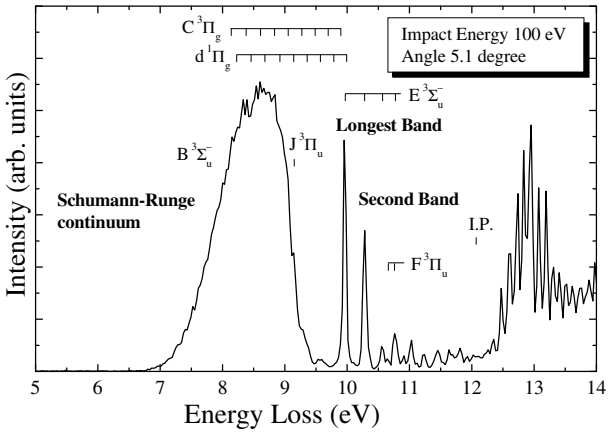


Fig. 3. Typical electron energy loss spectrum, $\Delta E \sim 4.9$–14.4 eV, for electronic state excitation in O_2.

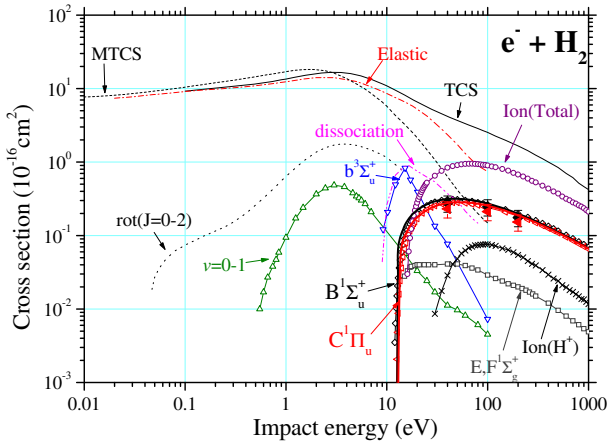


Fig. 2. (Color online) Recommended e^- + H_2 cross section data set from Itikawa [7], updated to include the recent ICSs from Sophia University [14] and with the corresponding BE*f*-scaling results.

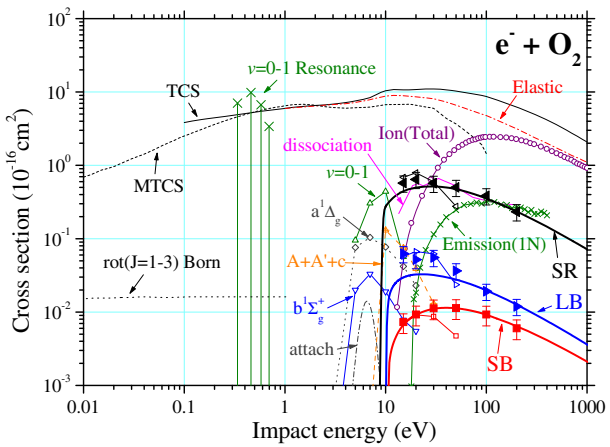


Fig. 4. (Color online) Recommended e^- + O_2 cross section data set from Itikawa [8], updated to include the recent ICSs from Sophia University [15] and with the corresponding BE*f*-scaling results.

method, the calculated ICSs agree well with the experimental results over the entire common energy range for the $B^1\Sigma_u^+$ and $C^1\Pi_u$ excitation states, which are consistent with the previously recommended data. Full details, including a discussion and references, can be found in references [1,7,14].

3.2 O_2

For O_2, EELS spectra were measured at various incident energies in the range from 15 to 200 eV, for electron scattering angles ranging from 2° to 130°. In Figure 3 a typical EELS spectrum covering the $B^3\Sigma_u^-$ (the Schumann-Runge continuum: SR), the $E^3\Sigma_u^-$ ($\nu' = 0$) (longest band: LB) and the $E^3\Sigma_u^-$ ($\nu'' = 1$) (second band: SB) electronic excitation states is plotted over the energy-loss range from 4.9 eV to 14.4 eV at $E_0 = 100$ eV and $\theta = 5.1°$. It is well known that the SR continuum, as well as the LB and the SB states in O_2 suffer (to varying degrees) from the effects of extensive Rydberg-Valence interactions. In particular, it is generally considered that these Rydberg-Valence perturbations lead to an anomalous (non-Franck-Condon) behaviour in the vibrational intensity distribution of the $E^3\Sigma_u^-$ state. Hence, the LB and SB transitions actually arise as excitations from the ground molecular state of O_2 to the lowest two members of the vibrational progression in the excited $E^3\Sigma_u^-$ electronic state.

The updated cross section data set for O_2 in the energy range from threshold to 1000 eV is shown in Figure 4, with our ICSs for the SR continuum, the LB and the SB electronic excitation states assessed against our BE*f*-scaling computations.

For the SR continuum transition, the absolute values of our experimental ICSs are largely consistent with the higher (>100 eV) energy experimental data and the calculations available. Our data in the lower (<100 eV) energy range is also appropriate for the trend of the cross section, though this agreement is not as good when compared to previous results from other groups [15]. Our experimental SR-continuum ICSs are systematically somewhat higher in magnitude than our corresponding BE*f*-scaling results for energies less than about 100 eV, suggesting contamination

from the accidentally degenerate or near-degenerate state of $C^3\Pi_g$ and $d^1\Pi_g$ classification.

For the LB, there is significantly less measured data available and no other theory against which we can compare the recent Sophia University results. The breakdown in the applicability of the BE*f*-scaling method emerged here due to the well-known strong Rydberg-Valence perturbations that affect the $E^3\Sigma_u^-$ ($\nu' = 0$) state in O_2 [15].

For the SB, the agreement with our experimental data is excellent throughout the energy range and through our assessment, by applying the BE*f*-scaling method, a more reliable ICS can now be provided.

Full details, including a discussion and references, can be found in references [1,8,15].

3.3 CO

For CO, EELS spectra were measured at various incident energies in the range 20 to 200 eV, for the scattered electron angle range from 3.5° to 130°. A typical EELS spectrum is plotted for the $A^1\Pi$, $C^1\Sigma^+ + c^3\Pi$ and $E^1\Pi$ states in Figure 5. Here the $A^1\Pi$ state consists of the vibrationally resolved ($\nu = 0{-}8$) sublevels, while the $c^3\Pi$ electronic state is almost degenerate in energy with the $C^1\Sigma^+$ state. As there was no possibility, with the current energy resolution, of uniquely resolving those states, no attempt was made to do so. The $A^1\Pi$ electronic state is a strong optically allowed transition from the ground state and is well separated from other excited electronic states in the spectrum of CO. It is thus an excellent model to test the BE*f*-scaling approach.

The updated cross section data set for CO is shown in Figure 6, with our ICSs for the $A^1\Pi$, $C^1\Sigma^+ + c^3\Pi$ and $E^1\Pi$ electronic excitation states benchmarked against our own BE*f*-scaling computation for the wide energy range from threshold to 1000 eV.

For the $A^1\Pi$ state, excellent agreement was found between the experimental and BE*f*-scaled ICS. For the $C^1\Sigma^+ + c^3\Pi$ states, at the higher (>30 eV) energies, the $C^1\Sigma^+$ BE*f*-scaling calculation is in very good agreement with our experimental data and most of the other experimental data and agrees at least to within the stated uncertainties for the ICSs [17]. At the lower (<30 eV) energies, however, the available experimental ICSs tend to be somewhat higher in magnitude than the BE*f*-scaling result. This discrepancy is due to the contribution from the $c^3\Pi$ state that cannot be experimentally resolved. For the $E^1\Pi$ state, there is excellent agreement between the experimental data and the BE*f*-scaled result. Full details, including a discussion and references, can be found in references [3–5,16,17].

3.4 CO_2

For CO_2, EELS spectra were measured at various incident energies in the range 30 to 200 eV, for the scattered electron angle range from 3.5° to 130°. A typical EELS spectrum covering the $^1\Sigma_u^+$ and $^1\Pi_u$ excitation states is displayed in Figure 7, over the energy-loss range from ~6 eV to 14 eV, at $E_0 = 100$ eV and $\theta = 15°$.

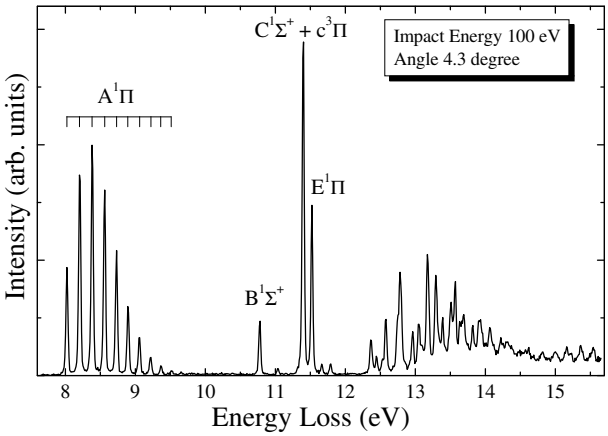


Fig. 5. Typical electron energy loss spectrum, $\Delta E \sim 7.5$–15.5 eV, for electronic state excitation in CO.

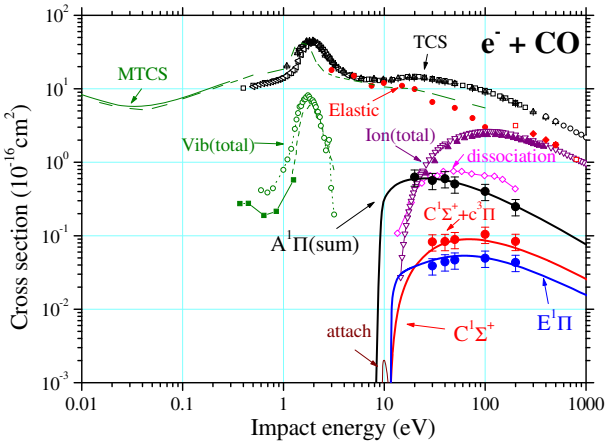


Fig. 6. (Color online) Recommended e^- + CO cross section data set from Zecca and colleagues [3–5], updated to include the recent ICSs from Sophia University [16,17] and with the corresponding BE*f*-scaling results.

The updated cross section data set for CO_2 is shown in Figure 8, with our ICSs for the $^1\Sigma_u^+$ and $^1\Pi_u$ electronic excitation states assessed against our BE*f*-scaling computation for the wide energy range from threshold to 1000 eV. We note that we derived our theoretical ICSs from the work of Beunker et al. [32]. Within the various stated uncertainties, the level of agreement is satisfactory between the experimental ICSs and those calculated within the BE*f*-scaling method for both the $^1\Sigma_u^+$ and the $^1\Pi_u$ electronic excitation states over the entire common energy range. Full details, including a discussion and references, can be found in references [1,9,18].

3.5 N_2O

For N_2O, EELS spectra were measured at various incident energies in the range 15 to 200 eV, for the scattered electron angle range from 2° to 130°. A typical EELS

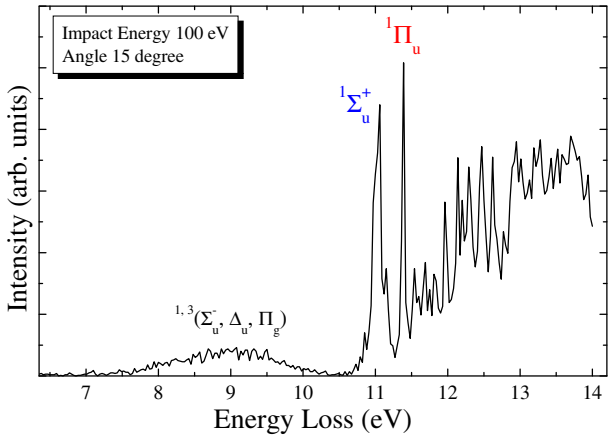


Fig. 7. (Color online) Typical electron energy loss spectrum, $\Delta E \sim 6.0-14$ eV, for electronic state excitation in CO_2.

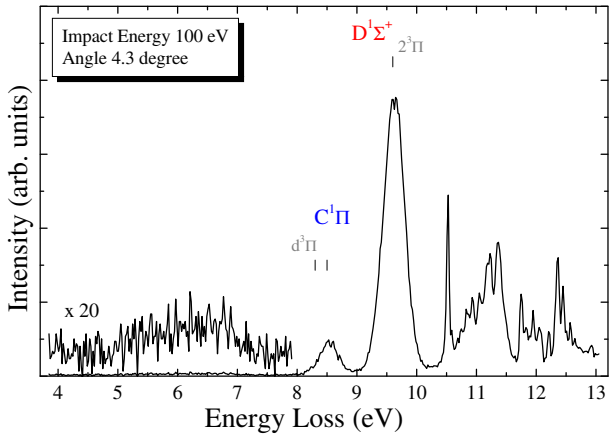


Fig. 9. (Color online) Typical electron energy loss spectrum, $\Delta E \sim 4-13$ eV, for electronic state excitation in N_2O.

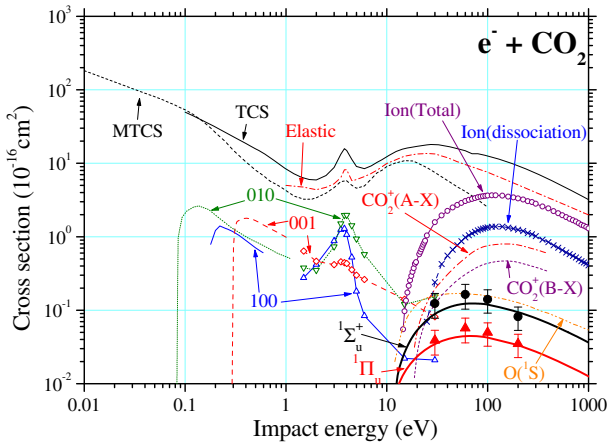


Fig. 8. (Color online) Recommended e^- + CO_2 cross section data set from Itikawa [9], updated to include the recent ICSs from Sophia University [18] and with the corresponding BE*f*-scaling results.

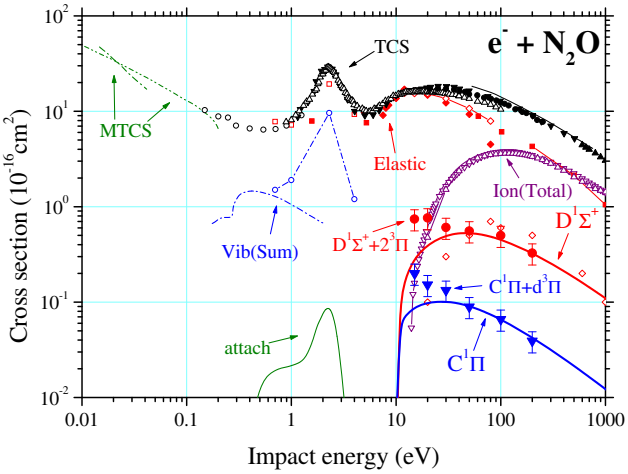


Fig. 10. (Color online) Recommended e^- + N_2O cross section data set from Zecca et al. [3–5], updated to include the recent ICSs from Sophia University [20] and with the corresponding BE*f*-scaling results.

spectrum, that includes the $C^1\Pi$ and $D^1\Sigma^+$ electronic excitation states, is plotted in Figure 9 over the energy-loss range from 4 eV to 13 eV and at $E_0 = 100$ eV and $\theta = 4.3°$.

The updated cross section data set for N_2O is shown in Figure 10, with our ICSs for the $C^1\Pi$ and $D^1\Sigma^+$ electronic excitation states assessed against our BE*f*-scaling computation for the wide energy range from threshold to 1000 eV. Note that the BE*f*-scaling ICSs were obtained using data taken from references [33,34]. The ICSs show common features for both the $C^1\Pi$ and $D^1\Sigma^+$ electronic excitation states, which are considerably lower in magnitude than other molecules as discussed in this report. At higher (>30 eV) energy, the ICSs agree well with the BE*f*-scaling results, but they progressively deviate from one another below 30 eV and towards the threshold [20]. This breakdown [20] in the applicability of the BE*f*-scaling method is due to the overlapping of a $d^3\Pi$ state with the $C^1\Pi$ state at 8.3 eV (nearly degenerate) and a $2^3\Pi$ state with the $D^1\Sigma^+$ at 9.6 eV (nearly degenerate), respectively. Full details, including a discussion and references, can be found in references [3–5,20].

3.6 H_2O

For H_2O, EELS spectra were measured at various incident energies in the range from 20 to 200 eV, for the scattered electron angle range from 3.5° to 130°. In Figure 11 a typical EELS spectrum is plotted for the relevant excitation states, over the energy-loss range from 6 eV to 11 eV and at $E_0 = 200$ eV and $\theta = 3°$. Note that this is an important channel for the production of the OH radical. However, even for this very important molecule there are currently only scanty experimental results for its electronic excitations.

The updated cross section data set for H_2O is shown in Figure 12, with our ICSs for the $\tilde{A}^1B_1$ electronic excitation state assessed against our BE*f*-scaling computation for the wide energy range from threshold to 1000 eV.

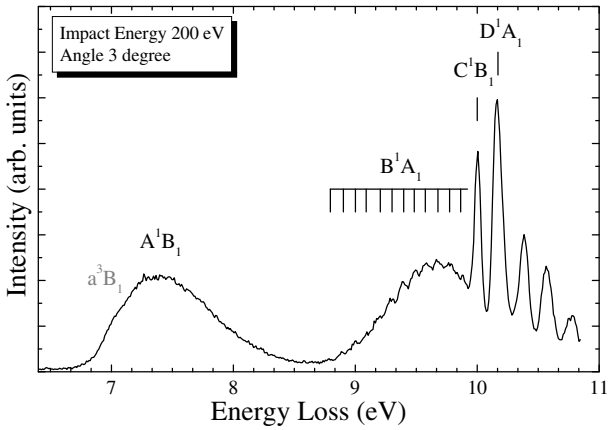


Fig. 11. Typical electron energy loss spectrum, $\Delta E \sim$ 6–11 eV, for electronic state excitation in H_2O.

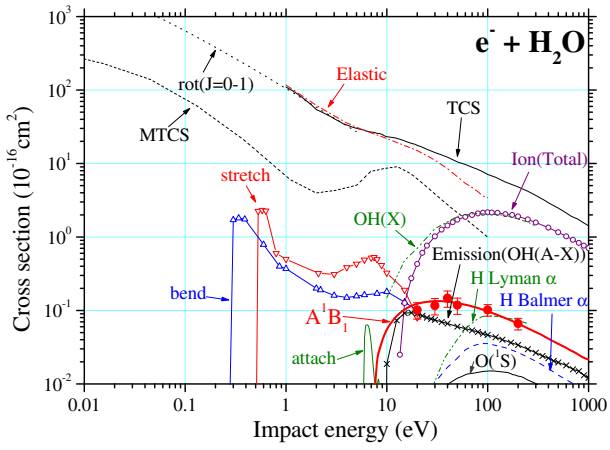


Fig. 12. (Color online) Recommended $e^- + H_2O$ cross section data set from Itikawa and Mason [10], updated to include the $\tilde{A}^1B_1$ ICSs from Flinders University [19] and with the corresponding BE*f*-scaling results.

Though the low-lying $^3B_1$ state overlaps with the $\tilde{A}^1B_1$ state, the agreement is very good and no breakdown was observed in the applicability of the BE*f*-scaling method. Full details, including a discussion and references, can be found in references [1,10,19].

4 Conclusions

Those who use cross section databases for modelling of phenomena, should note that any recommended values of cross section data might need to be updated due to the development of new experimental techniques and theoretical methods. This is particularly relevant as to this day there remain very few molecular cross sections that can be considered to have been benchmarked. In validating a scaling method for the Born cross sections of molecules, the present assessment shows that: (1) the BE*f*-scaling method is effective with respect to describing a single channel for optically allowed transitions. (2) For experimentally unresolved bands, composed of an optically allowed and an optically forbidden state, one might also deduce the cross sections for the relevant optically forbidden state by using the BE*f*-scaling ICSs with the experimental data. The current BE*f*-scaling results therefore provide a useful and accurate guideline to develop the databases needed for scientists wishing to model applications.

Finally, we note that while we have concentrated here on absolute electronic-state cross sections, those for other processes including elastic scattering are also crucial. The task to providing benchmark data for so many relevant species is massive and can in our view only be achieved through strong international collaboration.

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